# BL-29 / BL-31 verification follow-up

This follow-up reviews merged PR #50 (`577489b`) and verifies the changes at `daebfc1`, including main `60cdba8` (PR #52). The retained JSON records actual execution timestamps and source fingerprints. All records, accounts, emails and backups used for verification are synthetic and isolated from `attendance_dev`.

## Code changes and defect retests

- **Dashboard calculations:** derive subject and pooled shortages from one grouped attendance query.

This removes a second scan and keeps both metrics on the same database snapshot. Counts remain live; faculty scope, exact threshold comparisons and unequal subject weights are preserved.

- **Attendance notifications:** insert a threshold-crossing batch together and reload committed

recipients once. Each student still receives an individual email after the attendance transaction commits. A failed recipient does not stop the others; rollback emits no email. A 60-student submission uses **30 queries instead of 148** in the retained sequential profiles. Notification insert/read queries drop from **120 to 2**; each staff dashboard uses **11 queries instead of 12**.

- **Report validation / #51:**

UT*092* in the refined test plan required a report scope and date range, but `main` generated unfiltered downloads. HTML, CSV and PDF now require both dates and an explicit scope. The initial page only shows the filter form. Users can choose a student, class-and-subject, class/section, or explicitly select all their authorized classes. Invalid/empty input produces no download or export audit event. Section selection cannot reveal another teacher's subjects in the same section.

- **Evidence guards:** shortened runs cannot claim full acceptance; at least 30 rounds and three

marking/report repetitions are needed. Altered workloads and seed manifests from another plan, database or application revision are rejected. Targets were not weakened.

The report-query changes in the benchmark supply the now-required explicit scope/date parameters. Background report requests still include all of the faculty member's authorized records across the whole seeded calendar through the benchmark day. The foreground semester export selects F000's section, which contains exactly the same five assignments, 300 student/subject rows and 22,500 marks as before. These changes do not reduce the measured dataset or replace generated reports with an empty landing page.

## Local performance results

Environment: macOS arm64, 10 logical CPUs, Python 3.14.6, Django 5.2.18, PostgreSQL 18.4; 128 MB shared buffers and 4 MB work memory. The complete frozen BL29-v1 workload uses one Python process, 50 synchronized threads, real Django middleware, persisted OTP-verified sessions and CSRF. Email callbacks run synchronously with the in-memory mail backend. No other verification workload ran during the timed baseline or final benchmark.

| **Check** | **Baseline** `577489b` | **Final** `daebfc1` | **Required target / result** |
|---|---|---|---|
| Common requests within 2 s | 1,500/1,500 | 1,500/1,500 | At least 95%; passed |
| Common p95 | 0.220 s | 0.186 s | Below 2 s |
| Common maximum | 0.265 s | 0.212 s | Recorded |
| Slowest 60-student submission | 0.306 s | 0.286 s | Each of three at most 3 s; passed |
| Slowest semester export | 0.200 s | 0.156 s | Each of three at most 10 s; passed |
| Functional requests | 1,800/1,800 | 1,800/1,800 | No failures; passed |
| Peak in-flight requests | 50 | 50 | 50 authenticated users; passed |

The baseline already passes on this Mac. These measurements therefore **do not demonstrate that the earlier Windows latency failure has been resolved**, nor do they establish deployment latency. The query reduction is reproducible independently of host speed; small timing differences are not a guaranteed speedup. The final run starts with 1,000 students, 60 faculty and 750,000 marks, ends with 750,180 marks in 12,753 sessions, and verifies 60 first-crossing notifications/emails with no duplicates on the subsequent two submissions. All PT*01-PT*04 checks pass within the declared local boundary.

Evidence: baseline, final plan, final seed, final per-request outcomes, profile before, profile after. The profiles cover the query optimization at `5c956ae`; the final benchmark includes the report fix. Profiles are sequential diagnostics, not concurrent acceptance timings.

### Independent GitHub verification

Workflow run 41153839750 **passed** on Ubuntu at `27b5329`, before the subsequent PR #52 navigation merge. It independently ran the complete suite (**473 passed, 3 Windows-only skips**) and all four performance checks against PostgreSQL 18.6, Python 3.14.9 and four logical CPUs. All 1,800 requests succeeded; common p95 was **0.901 s**, the slowest marking submission **1.285 s**, and the slowest semester export **0.964 s**. All 1,500 common requests met the two-second target. These remain in-process test-stack measurements, not external HTTP/deployment or Windows measurements.

The downloaded plan, seed, per-request results and per-test results are retained locally so the evidence survives GitHub artifact expiry. The integrated Mac run above includes the later navigation merge. The latest standard CI checks are attached to PR #53.

## Integration, security and regression

The final suite discovered **476 tests: 473 passed, 0 failures/errors, 3 skipped**, in **80.635 seconds**. It used the application's default password hashers. The three skips require native Windows ACL/junction APIs; the existing PR #50 evidence records their earlier Windows execution. No new Windows execution is claimed here. Ruff lint/format, Django system checks, migration drift, dependency consistency and diff whitespace checks all passed.

The retained per-test outcomes provide actual names, statuses and durations. Static checks retain commands and exit codes.

| Refined test-plan area | Executed coverage |
|---|---|
| UC-01 / BL-33 / BL-34 | Real password and OTP setup/login, credential delivery, forced password change, replay/expiry, lockouts, recovery, reset/session revocation |
| UC-02-05 | Student/faculty/subject/section provisioning and edits, enrollment/assignment validation, scope denial, preservation and deletion policies |
| UC-06-07 | Complete marking, duplicate/partial/concurrent saves, exact calculations, correction-window boundaries, Admin override reasons and atomic audit |
| UC-08 / UC-12 | Own-student views, filters, zero-class states, pooled totals, role dashboards, logout and expired sessions |
| UC-09 / UC-11 | Explicit scope/dates, section authorization, report denominator consistency, private/audited/formula-safe exports, empty/invalid/unsupported downloads, history/search |
| UC-10 | Exact threshold boundaries, first crossing/recovery/recrossing, private batch delivery, rollback, transport failure and weekly faculty notifications |
| UC-13 | Genuine encrypted backup/restore, integrity rejection, rollback, preserved audit, session revocation, maintenance/concurrency gates and readiness failures |

Five new report methods cover the missing UT*09*2 behavior and section scope. Five query/batch regression methods cover pooled counts, private delivery, current recipient addresses, deleted notifications, per-recipient failure and rollback. Four additional evidence tests reject invalid workloads/manifests and false full-sampling claims. Existing integration journeys now pass explicit report selections while retaining their original data/authorization assertions.

The Word test plan's UT*06*4 uses rejection language for all duplicate submissions. BL-13 intentionally supports an identical retry without creating records; a changed retry is rejected and directed to correction. The existing submission-safety tests verify both behaviors. This report does not claim that every literal manual script was performed in a browser; browser/device/usability acceptance remains #30.

## Availability and completion boundary

The owner confirmed that there is **no deployed system or existing uptime history**. A real local `/health/` probe returned HTTP 200; the monthly summary correctly remains **pending**, with **0% institutional-hours coverage**. The local probe occurred outside the institutional window and cannot establish monthly uptime.

| Issue | Verified now | Still needed before full closure |
|---|---|---|
| #29 | Frozen workload, representative capacity, all local performance checks, availability/recovery tooling and regression coverage | Intended/representative deployment test and a fully observed month meeting 99% availability |
| #31 | Updated code, scoped report fix and full local integration/security/regression evidence | Merge/review the follow-up and resolve agreed release-blocking performance acceptance; no blanket release approval is claimed |
| #49 | Profiled query costs, reduced database work, passing before/after Mac benchmarks | Retest the original Windows configuration or an agreed representative deployment; Mac success does not invalidate the earlier failure |
| #51 | Validation and authorization fix with passing regression tests | Merge the follow-up |

Use `verification/README.md` to repeat the benchmark on a fresh disposable database and to configure real health monitoring after deployment. The target, measurement boundary and missing observations must remain explicit. The manifest records evidence hashes and confirms that JSON compaction preserved every measured value.